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Pink Gold: The Middle East & North Africa's Nuclear Powered Hydrogen Revolution

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Abstract

This review paper provides an analysis of the potential for pink hydrogen production in the Middle East, a region strategically positioned to leverage its growing nuclear energy sector for low-carbon hydrogen generation. The core objectives of this research are threefold:1. To conduct a comprehensive analysis of the current state of pink hydrogen production technologies. This involves evaluating their technical maturity, efficiency, and suitability for integration with the Middle East's existing and planned nuclear energy infrastructure, with a focus on comparing different electrolysis methods.2. To evaluate the techno-economic viability of pink hydrogen production within the Middle East's specific energy landscape. This entails assessing the levelized cost of hydrogen (LCOH), identifying economic challenges and opportunities, and considering factors such as infrastructure development and policy incentives.3. To identify the critical gaps that currently impede the development of a robust pink hydrogen economy in the Middle East. This includes an examination of infrastructure limitations, policy and regulatory shortcomings, and the crucial aspect of public acceptance of both nuclear energy and hydrogen technologies.

The research methodology employed a systematic literature review. This involved a comprehensive search and synthesis of data from a variety of sources, including peer-reviewed academic journals, industry reports from organizations like the IEA and IRENA, government publications outlining national energy strategies, and reputable online resources. Keywords such as "pink hydrogen," "nuclear hydrogen," "Middle East," "hydrogen strategy," and "electrolysis" were used to ensure a broad and relevant collection of information. The review highlights the Middle East's significant pink hydrogen potential. The region's substantial nuclear investments, notably in the UAE and planned expansions in Saudi Arabia and Egypt, provide a strong foundation. Potential applications include decarbonizing refining and steel production, and exports. However, key challenges must be addressed:1. Technological gaps (e.g., HTSE deployment).2. Economic considerations (e.g., high costs, competitiveness).3. Infrastructure limitations (e.g., storage, transport).4. Policy and regulatory gaps (e.g., clear frameworks).5. Public acceptance (e.g., nuclear energy concerns).

This review contributes novel information by providing a uniquely focused analysis of pink hydrogen within the Middle Eastern context. It moves beyond general global assessments of hydrogen to specifically examine the interplay between the region's nuclear energy ambitions and its potential to become a leader

in pink hydrogen production. This regionally specific approach, combined with a detailed identification of the key challenges and opportunities, offers valuable insights for policymakers, industry stakeholders, and researchers.

Introduction

The energy industry is undergoing rapid change. It is no longer feasible to continue delivering energy from the same polluting sources for very long, and climate change has become a serious hazard. To prevent more severe and permanent climate effects, fossil fuel consumption must be gradually reduced. Over the past 20 years, a number of clean energy sources, including solar, wind, and other renewable energy sources, have been developed. New problems that must be resolved arise as these energy sources begin to replace the outdated, polluting ones. Energy storage is one of the primary issues now being brought on by this energy shift. It is difficult to balance the population's energy needs with the amount of energy produced by renewable sources (Jain, 2024).

Another possible storage option that is being developed and may meet several needs is hydrogen, Figure 1 shows the Hydrogen (H_2) colours based on the source of energy. Hydrogen has evolved from niche industrial feedstock to cornerstone of global net-zero roadmaps (Razi and Dincer, 2022). One energy carrier that makes it possible to move energy from one location to another in a useful manner is hydrogen. By weight, it contains the most energy of any common fuel. It is worth noting that one kilogram of hydrogen contains roughly 33 kWh of energy, which is almost three times as much as gasoline (Yue *et al.*, 2021). Despite this, it has the lowest energy content by volume at atmospheric pressure—nearly four times less than gasoline. Clean, non-polluting, storable, versatile, and renewable are some of the benefits of hydrogen. It might be regarded as the 21st century's "ultimate energy." Construction, industry, electricity, and transportation are just a few of the industries that use it. On Earth, hydrogen is found naturally in gases, liquids, and solids as compounds with other elements. It can be mixed with carbon to create various compounds, such the hydrocarbons found in coal, petroleum, and natural gas, or it can be mixed with oxygen to create water. For hydrogen to be consumed on its own, it must be separated from the other elements. Naturally, producing and purifying hydrogen requires more energy than it yields when transformed into usable energy (Petrii, 2009). Conference of Parties 2029 (COP29) observed a significant push for Hydrogen production and usage. With the launch of COP29 Hydrogen Declaration, calling for collaboration to scale up hydrogen production has been established globally. COP 28 laid the groundwork for such a Declaration on a global stage by creating an awareness on phasing out fossil fuels while also reaffirming the Glasgow Agreement (COP26) in 2021 (Prasad 2025).

Grey Hydrogen	Hydrogen produced by fossil fuels, mostly natural gas and coal, causing CO_2 emissions
Blue Hydrogen	Hydrogen produced by fossil fuels in combination with CCS, reducing GHG emissions
Turquoise Hydrogen	Hydrogen produced via pyrolysis of fossil fuels, where the buy-product is solid carbon
Green Hydrogen	Hydrogen produced by electrolysis using electricity generated by renewable energy sources
Pink Hydrogen	Hydrogen produced by electrolysis using electricity generated by nuclear energy source
Gold Hydrogen	Hydrogen produced by electrolysis using grid electricity

Figure 1—Hydrogen Color Spectrum

Pink Hydrogen: A Global Perspective

Nuclear power presents a compelling pathway for clean hydrogen production, offering significant advantages in cost competitiveness, energy efficiency, and continuous baseload operation compared to renewable alternatives. A single 1,000 MW nuclear reactor can potentially produce over 150,000 metric tons of hydrogen annually, equivalent to about 15% of current U.S. hydrogen production. The steady electricity output from nuclear plants, combined with available process heat, creates ideal conditions for efficient electrolysis operations. High-capacity-factor nuclear plants (> 90 %) are uniquely suited to feed electrolysers continuously, mitigating degradation associated with renewable intermittency. High Temperature Steam Electrolysis (HTSE) can also utilize reactor waste-heat steam, pushing electric-to-hydrogen efficiency from the 54–60 kWh kg^{-1} (alkaline/PEM) range down to 45 kWh kg^{-1} (Bass et al., 2022).

Pink hydrogen is a proven concept around the globe and has been implemented in certain cases. Understanding the technology is crucial to further determine the feasibility under comprehensive techno-economic assessment. A typical integrated nuclear powered hydrogen production plant would include a viable nuclear energy plant, water treatment system, electrolyser, heat integration system and hydrogen processing and storage systems, see Figure 2.

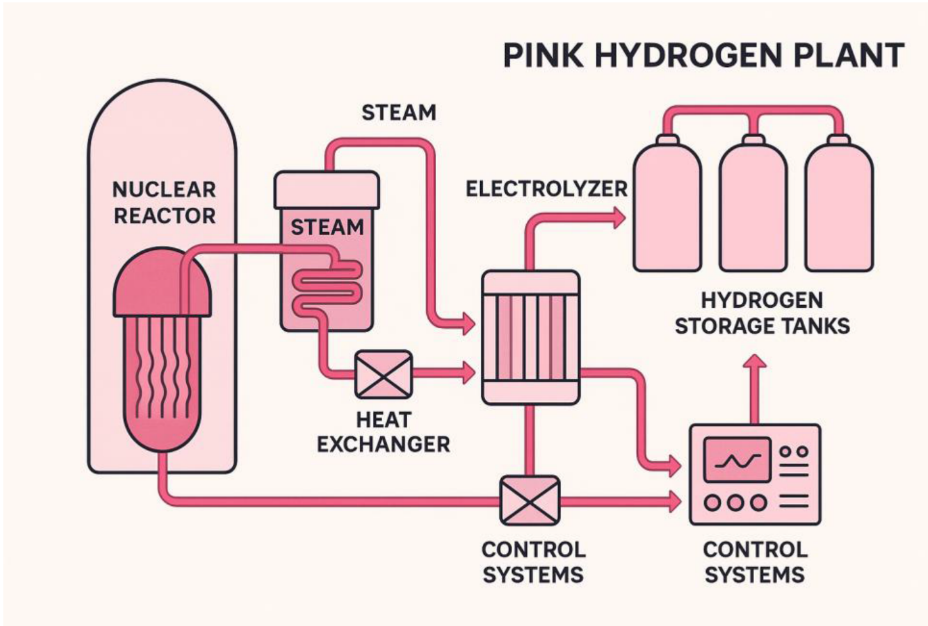

Figure 2—Integrated Nuclear Powered Hydrogen Production plant – Pink Hydrogen

Water Treatment Systems

All electrolyser types require highly purified water to prevent electrode degradation and maintain efficiency. Water treatment systems include reverse osmosis units, deionizers, and multi-stage purification systems to achieve the required water purity levels. The water treatment capacity must match the electrolyser's consumption requirements (Husaini et al. 2025; Marie Esposito et al. 2024; Sgarbi et al. 2025).

Heat Integration Systems

Heat integration represents a critical advantage of nuclear-hydrogen coupling, particularly for high-temperature applications.

- Heat Exchanger Networks: Advanced heat exchanger designs, including printed circuit heat exchangers (PCHEs), enable efficient thermal energy transfer from nuclear reactors to hydrogen production processes. These compact, high-efficiency heat exchangers can handle the temperature and pressure requirements of nuclear-hydrogen integration while maintaining safety isolation between nuclear and hydrogen systems (Kupecki et al. 2024).
- Steam Distribution Systems: Steam systems transport thermal energy from nuclear reactors to electrolysis processes, particularly for SOEC applications. High-temperature steam piping, pressure control systems, and thermal expansion accommodation are essential components. Steam conditions must be carefully matched to electrolyser requirements (Knighton et al. 2021).
- Thermal Management Systems: Sophisticated thermal management maintains optimal temperatures throughout the hydrogen production process. This includes heat recovery systems that capture waste heat from electrolysis for process preheating, improving overall system efficiency. Temperature control systems ensure safe and efficient operation across varying load conditions (Marie Esposito et al. 2024).
- Process Heat Recovery: Heat recovery systems capture thermal energy from various process streams to improve overall plant efficiency. Recuperative heat exchangers preheat feedwater using waste heat from hydrogen and oxygen product streams, reducing the overall energy requirement (Kupecki et al. 2024).

Hydrogen Processing and Storage Systems

The hydrogen processing section purifies, compresses, and stores the produced hydrogen for distribution and use. The following systems are integral part of a working hydrogen processing plant.

- Gas Separation Systems: Electrolysis produces hydrogen and oxygen that must be separated and purified. Gas-liquid separators remove entrained water, while pressure swing adsorption (PSA) systems can provide additional purification if required. Advanced separation technologies ensure high-purity hydrogen production. (Diaz-Pescador *et al.*, 2025; Marie Esposito *et al.*, 2024). Compression Systems: Hydrogen compression equipment increases gas pressure for storage and transportation. Multi-stage compression with intercooling manages the temperature rise associated with hydrogen compression. Compressor selection depends on the required delivery pressure and flow rates (Marie Esposito *et al.*, 2024).
- Storage Systems: Hydrogen storage includes both short-term buffer storage and longer-term storage systems. High-pressure gas storage tanks, potentially utilizing underground salt caverns for large-scale storage, provide inventory management capability. Cryogenic liquid hydrogen storage may be included for some applications requiring higher energy density storage (Brauns and Turek, 2020).
- Purification Systems: Additional purification equipment ensures hydrogen meets quality specifications for various end-use applications. This may include catalytic purifiers, molecular sieves, or other specialized purification technologies depending on the required hydrogen purity level (Król *et al.*, 2024).
- Control and Instrumentation Systems: Advanced control systems coordinate operation between nuclear and hydrogen production systems. This includes distributed control systems (DCS), safety instrumented systems (SIS), and process monitoring equipment. Integration requires careful coordination to maintain both nuclear safety and hydrogen production efficiency (Flamm *et al.*, 2021).
- Safety Systems: Comprehensive safety systems address both nuclear and hydrogen-specific hazards. Fire protection systems, hydrogen leak detection, ventilation systems, and emergency shutdown capabilities are essential. Gas detection systems monitor for hydrogen leaks, while fire suppression systems are designed for hydrogen fire scenarios (Brauns and Turek, 2020; Ireland *et al.*, 2025).
- Utilities and Support Systems: Essential utilities include nitrogen systems for purging and inerting, instrument air systems for valve actuation, and cooling water systems for process cooling. Electrical support systems provide uninterruptible power for critical instrumentation and control systems (Kupecki *et al.*, 2024).
- Waste Management Systems: Integrated waste management handles both conventional industrial waste and any radioactive waste from nuclear operations. Water treatment systems manage process wastewater, while specialized systems handle any contaminated materials according to nuclear regulatory requirements (Marie Esposito *et al.*, 2024; Sgarbi *et al.*, 2025).

System Integration Considerations

- Nuclear-Hydrogen Interface: The interface between nuclear and hydrogen systems requires careful design to maintain nuclear safety while enabling efficient hydrogen production. This includes physical separation systems, redundant isolation capabilities, and coordinated control systems that can respond to both normal and emergency conditions (Kim *et al.*, 2024; Knighton *et al.*, 2021; Yoo *et al.*, 2025).
- Load Management: Advanced load management systems optimize the balance between electricity generation and hydrogen production based on grid demands and hydrogen market conditions. This

capability allows nuclear plants to provide grid stability services while maintaining continuous hydrogen production (Kim *et al.*, 2024; Knighton *et al.*, 2021; Yoo *et al.*, 2025).

- Regulatory and Licensing Integration: The combination of nuclear and hydrogen systems creates complex regulatory challenges requiring coordination between nuclear regulators and industrial safety authorities. Design must accommodate requirements from both regulatory domains while maintaining operational efficiency.

Various electrolysis technologies have emerged in the industry with matured TRL with proven integration with nuclear energy. The following are technologies that have proven integration with nuclear energy production systems.

High Temperature Steam Electrolysis (HTSE). High Temperature Steam Electrolysis (HTSE) using Solid Oxide Electrolysis Cells (SOECs) showcases the most promising technology for nuclear-hydrogen integration (Bass et al., 2022). Figure 3 represents the working of HTSE technology. This technology operates at temperatures between 600-900°C, making it ideal for coupling with nuclear reactors that can provide both electricity and high-temperature process heat (Hauch *et al.*, 2020).

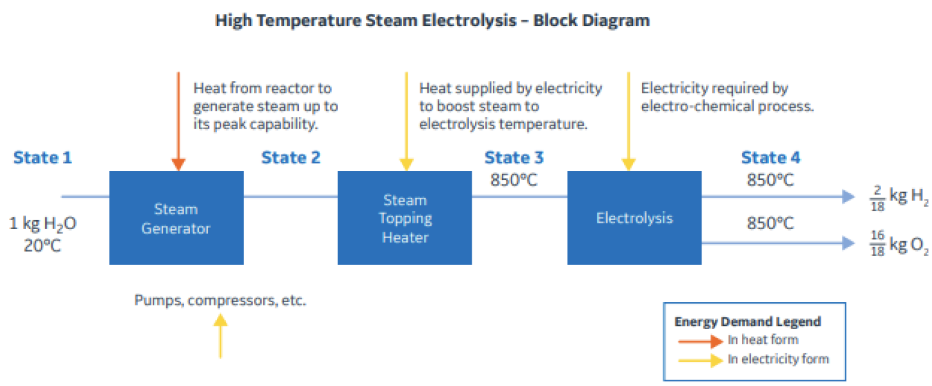

Figure 3—Block Diagram for HTSE (Bass et al., 2022)

SOECs demonstrate superior efficiency compared to low-temperature electrolysis methods. At 850°C, HTSE requires only 225 MJ of thermal energy per kg of hydrogen (64% efficient), compared to 350 MJ at 100°C (41% efficient). The high operating temperature reduces electrical energy requirements by approximately 30% compared to PEM electrolysis, as some energy is supplied as waste heat from the nuclear plant (GE Vernova, 2020; Idaho National Laboratory, 2014).

Recent advances in SOEC technology have been remarkable (Hauch et al., 2020). The initial electrochemical performance of state-of-the-art SOEC single cells has more than doubled over the past 10-15 years, while long-term durability has been improved by a factor of approximately 100 (Hauch et al., 2020). These improvements have led to a hundredfold increase in gas production capacity within the past decade (Hauch et al., 2020).

HTSE technology can be integrated with various nuclear reactor types. Light Water Reactors (LWRs) can provide steam at their outlet temperatures, while advanced reactors like High Temperature Gas-cooled Reactors (HTGRs) and Small Modular Reactors (SMRs) offer even better integration possibilities (GE Vernova, 2020; UKNNL, 2024; Rolls-Royce SMR, 2023). Studies show that a 600 MWth reactor could produce approximately 85 million SCFD (2.5 kg/s) of hydrogen (Department of Energy, 2014). The Idaho National Laboratory (INL) has been the lead facility for HTSE research and development since 2003, conducting extensive testing including the Integrated Laboratory Scale (ILS) experiment with three modules of 240 cells each, initially producing 5.6 Nm^3/hr at 18 kW power (OECD, 2010).

Alkaline Water Electrolysis. Alkaline electrolysis (Figure 4) represents the most established and commercially mature technology for nuclear-hydrogen production (Department of Energy, 2025; IAEA, 2020). This technology operates at lower temperatures (typically 60-80°C) and requires only electrical energy from nuclear plants, making it compatible with existing nuclear power infrastructure (Ireland *et al.*, 2025).

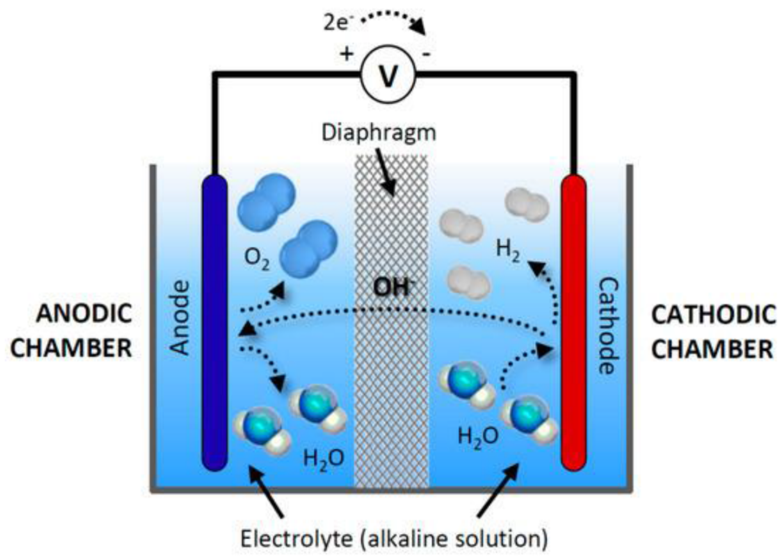

Figure 4—General Scheme & Operation of an alkaline electrolysis cell (Vera *et al.*, 2014)

Alkaline electrolysers use aqueous alkaline solutions (typically KOH or NaOH) as electrolytes and operate at atmospheric or slightly elevated pressures (Becker *et al.*, 2023). They offer several advantages including the use of non-precious metal catalysts, established manufacturing processes, and lower capital costs compared to other electrolysis technologies (ScienceDirect Topics, 2024; Department of Energy, 2025). Korean studies demonstrate successful integration of alkaline electrolysis with nuclear power. Nel, a Norwegian hydrogen firm, has partnered with Korea Hydro & Nuclear Power (KHNP) to explore combining alkaline electrolysis with nuclear power, building on KHNP's research into "pink hydrogen" production since 2022 (Currie, 2025).

Alkaline electrolysis represents the most mature technology for nuclear-hydrogen applications. These systems operate at relatively low temperatures (60-80°C) and can utilize standard electrical connections from nuclear plants. Alkaline electrolysers integrate primarily through electrical connections, typically using 480V or higher voltage systems from nuclear plant auxiliary power. The technology's robustness and lower temperature operation simplify integration compared to SOEC systems (Brauns and Turek, 2020). Swedish OKG's successful commercial operation demonstrates alkaline technology's reliability for continuous nuclear-powered operation. The system has operated successfully since 1992, providing hydrogen for reactor operations and now commercial supply (Dalton, 2024).

Polymer Electrolyte Membrane (PEM) Electrolysis. PEM electrolysis offers high efficiency and operational flexibility for nuclear integration (ScienceDirect, 2024; Bohrium, 2024). Operating at temperatures between 50-80°C, PEM electrolysers can respond rapidly to power fluctuations, making them suitable for load-following nuclear operations (NEI Magazine, 2025).

PEM electrolysers (Figure 5) demonstrate high current densities and can produce high-purity hydrogen (>99.9%) (Department of Energy, 2025). They operate at elevated pressures (up to 30 bar), reducing compression costs for hydrogen storage and transport (NEI Magazine, 2025). The technology uses proton-conducting polymer membranes and platinum-based catalysts (Department of Energy, 2025).

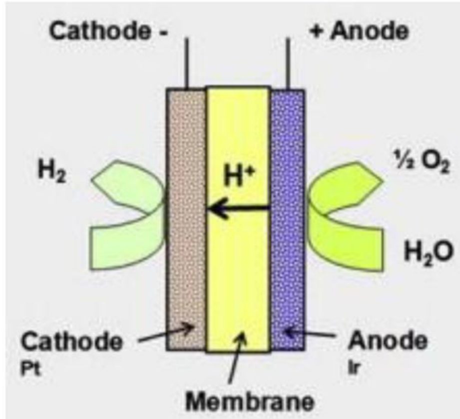

Figure 5—Polymer Electrolyser Membrane (Ahmad Kamaroddin *et al.*, 2021)

Recent research on integrating large-scale PEM electrolysis with the APR1400 nuclear power plant shows promising results for hydrogen production (Alabbadi and AlZahrani, 2024). The integrated system could produce hydrogen at an optimum efficiency of 53.5 %. The integrated system produces 651 t of H2/day at an energy efficiency of about 23%. The studies indicate that PEM electrolysis can effectively utilize nuclear electricity during off-peak demand periods. Pescador et al. (2025) had presented the outcomes from the impact assessment made on the integration of a 40 MW hydrogen production plant using PEM electrolyser technology with Rivne Nuclear Power plant in Ukraine. Hazardous points within the HPP were identified in the study using HAZID methodology, and its reliability was assessed. The study concluded that the highest frequency of vapor cloud explosion is in order of 10^{-7} per year following a 1% leak in the vessels further confirming the feasibility of such integration from a safety perspective (Diaz-Pescador *et al.*, 2025).

Nuclear-powered hydrogen production addresses several key challenges in the emerging hydrogen economy. The advantages for pink hydrogen can be grouped as follows:

1. Addressing climate change and Greenhouse Gas emissions: Unlike Blue hydrogen or black and grey hydrogen, Nuclear energy is the lowest carbon intense while being exceptionally reliable.
2. Cost Effectiveness: High hydrogen production is possible due to high technology matureness and due to high energy density from nuclear energy. Furthermore, the power plants offer a consistent and reliable electricity supply which is required for scaling Hydrogen production.

   a. Typical minimum energy requirement is between 39.4 to 55 kWh per kg of hydrogen for conventional electrolysis methodologies (Figure 6). However, when pink hydrogen is considered – the costs still hover in similar ranges but with increased efficiencies and reliable supply. For example, UK government studied compatibility between high temperature gas reactor (HTGR) and high temperature steam electrolysis (HTSE). Compared to other low-carbon energy production technologies that could also be coupled with hydrogen production plants, integration with nuclear energy is competitive.

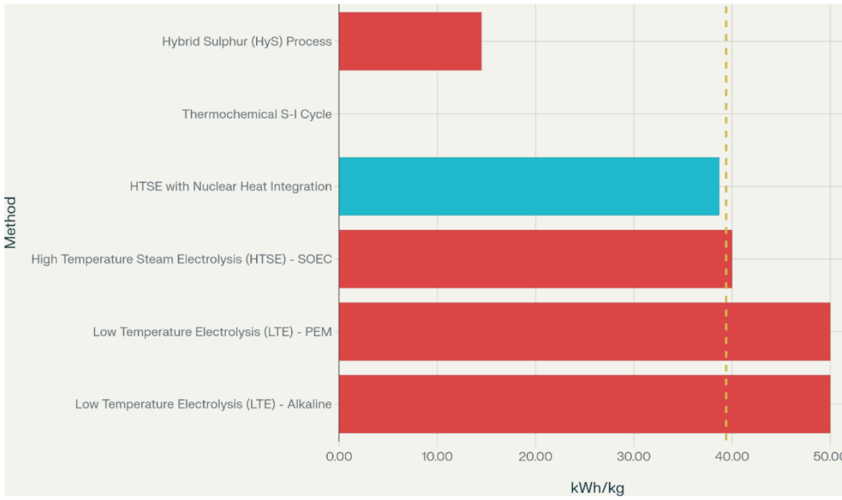

Figure 6—Hydrogen Production costs typical (UKNNL, 2024)

b. Nuclear-powered hydrogen production offers significant cost advantages over renewable alternatives, with costs ranging from \$1.0-3.5 per kg depending on the technology and reactor type which are recorded in the below table 1 (Arcos and Santos, 2023; DOE, 2024):

Table 1—Relative production costs based on literature (Arcos and Santos, 2023; DOE, 2024)

Hydrogen Production Method	Production Costs	Comparison with Green & Blue Hydrogen
Advanced thermochemical with high-temperature reactors	1.0-5.65 USD/kg	Pink hydrogen is 15 to 30 % cheaper than Blue Hydrogen (2-3.5 USD per kg) and Green Hydrogen (4.5 USD per kg)
Small Modular Reactor (SMR) with solid oxide electrolysis	2.0-3.7 USD /kg	
Nuclear HTSE (general range)	3.0-5.9 USD/kg	

3. Flexibility and Scalability: Nuclear power plants can be located close to industrial users of hydrogen which would in turn reduce transportation costs or can be located close to established infrastructure (e.g. Natural Gas pipeline networks)
4. Integration with existing Infrastructure: Additional CAPEX as mentioned above can be further optimized for transportation and existing nuclear powerplants can be leveraged for hydrogen production. Other possible integrations can be integrated with other hydrogen production methodologies which are also being studied.

Benchmarked cases around the globe include Nine Mile point nuclear station which is the first operational nuclear-powered hydrogen production system in the United States. Beginning operations in March 2023, this 1.25 MW demonstration uses a PEM electrolyser to produce 560 kg/D. The system operates at 13.8 kV, drawing power directly from the plant's electrical bus system (Du *et al.*, 2025). The project's key innovation lies in its integration approach, where hydrogen production replaces previously trucked-in fossil fuel-derived hydrogen used for reactor cooling operations. The PEM electrolyser system includes containerized units with integrated hydrogen storage, leveraging the plant's existing hydrogen infrastructure. This demonstration has proven the technical feasibility of nuclear-hydrogen coupling and provided valuable operational data for future scaling (Aminaho *et al.*, 2025). The Palo Verde project, supported by \$20 million in federal funding, focuses on a power-to-power demonstration concept. The system will produce hydrogen through electrolysis and subsequently use the stored hydrogen to generate electricity during peak demand periods. With 6 tonnes of hydrogen storage capacity, the project aims to produce 200 MWh of electricity during high-demand periods (Soja *et al.*, 2023).

Another project in US, the Davis-Besse project, funded with $8 million from the Department of Energy, represents a significant scaling effort targeting 100 metric tons of hydrogen production daily. The facility employs low-temperature alkaline electrolysis with a planned 2 MW installation. Strategic site selection considered proximity to industrial hydrogen consumers, particularly in Ohio's manufacturing and transportation sectors (IAEA, 2023).The project design incorporates steam integration from the nuclear plant to improve electrolysis efficiency. Unlike the Nine Mile Point project's focus on internal consumption, Davis-Besse targets external hydrogen markets, including potential fuel cell vehicle applications and industrial processes.

Xcel Energy's Prairie Island project stands out for its adoption of high-temperature SOEC technology in partnership with Bloom Energy (Knighton *et al.*, 2021). The 240 kW demonstration system utilizes both electricity and steam from the nuclear plant, achieving approximately 30% higher efficiency compared to conventional cold electrolysis. The SOEC system operates at elevated temperatures (650-850°C), allowing it to utilize waste heat from the nuclear plant. This thermal integration represents a key advancement in nuclear-hydrogen coupling, demonstrating how process heat can significantly improve overall system efficiency (Knighton *et al.*, 2021).

The Japan Atomic Energy Agency (JAEA) and Mitsubishi Heavy Industries (MHI) (Figure 7) collaboration at the HTTR facility focuses on high-temperature hydrogen production using steam electrolysis. The HTGR reactor's 950°C (Figure 8) outlet temperature enables direct thermal coupling with electrolysis systems. This project represents advancement in thermochemical hydrogen production cycles, including the sulfur-iodine cycle and high-temperature steam electrolysis. The demonstration aims to validate mass production of carbon-free hydrogen using ultra-high temperature nuclear heat (Sun *et al.*, 2010).

Figure 7—Hydrogen Production Hub (Industries, 2025)

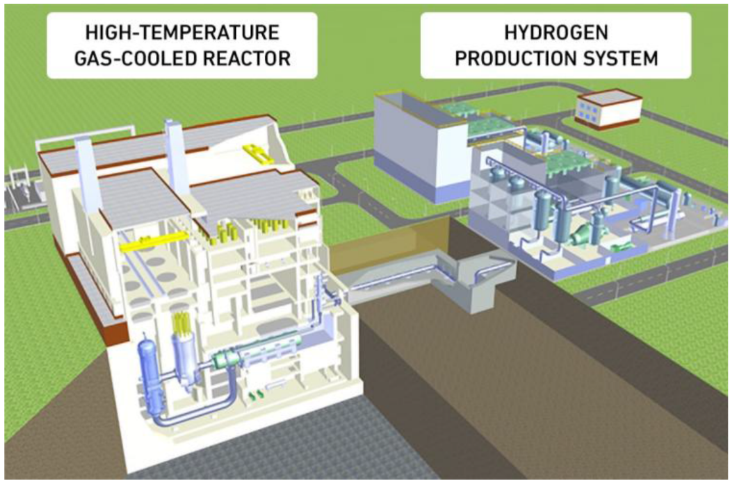

Figure 8—HTGR Layout (Willige, 2022)

Pink Hydrogen in MENA

Middle east is at the point of crossroad that may define its future in the energy market. Western countries have already opted for more innovative renewable energy sources while also developing efficient mannerisms to exploit non-renewable resources. Such proven technologies utilized globally can be implemented in the ever-growing Middle East and can showcase its importance in both technology and energy globally.

The most suited hydrogen production method for the Middle East is from fossil fuels and natural gas (Razi and Dincer, 2022). However, there are other avenues that can assist and complement the Middle East's future vision. While "green" (renewable electrolysis) and "blue" (fossil with CCS) hydrogen dominate investment pipelines, *pink hydrogen*—produced with zero-carbon nuclear electricity—remains comparatively under-examined, especially across the MENA region. Although nuclear power has been a topic of discussion over the past decades, utilizing it as a driver to produce another form of energy has not been under focus yet. Nonetheless, almost 400 nuclear reactors are operational worldwide (Fernández-Arias *et al.*, 2020). In the field of nuclear power plants under construction, more than 50 are in this situation, with China leading the list of countries that are in the expansion phase of new nuclear power plants to meet the growing demand for energy. Along with US, countries like China, India, Turkey, and the United Arab Emirates are also building new nuclear power plants. Introduction of pink hydrogen to MENA region in this period of time would be adversely benefit all the major players, while ushering the region into a completely new segment of the energy market. With the current development plans and each nation visualizing its diversified energy grid through Nuclear and Renewables, Hydrogen would equally have a better and profitable future in the region.

Unlike solar and wind, nuclear energy is reliable and consistent and cheaper. Hong, Qvsit and Brook (2018) studied a potential replacement of nuclear fission reactors for energy production by solar and wind. However, the results concluded that to reliably meet energy demand each hour of the year, the annual spending on electricity systems will be five times higher than the nuclear option. Furthermore, interestingly the greenhouse emissions would increase greenhouse emissions and in water-stressed MENA, co-location with coastal desalination reduces feed-water penalties (~9 L de-ionized water kg^{-1} H_2). Affordability has been under debate since two decades when studying the feasibility of renewables, nuclear on the other hand falls short on safety and public acceptance (Hong *et al.*, 2018). Nevertheless, readiness differs markedly: capital costs for new reactors range from 4 000 USD kW^{-1} (SMR optimistic) to > 6 500 USD kW^{-1} (large Gen-III+ builds); weighted-average cost of capital (WACC) spans 6–10 %; and public acceptance fluctuates (Sustainability 2023).

Existing Infrastructure

The following table 2 provides an overview of the current and planned infrastructure relevant to pink hydrogen production across key MENA countries. It summarizes each country's baseline industrial assets, initiatives for hydrogen and clean energy, the status of nuclear power plants and electrolyser deployment, as well as prospective projects and strategic direction. This comparative perspective highlights the region's varying readiness to support pink hydrogen—produced using nuclear energy—by detailing existing facilities, notable investments, and efforts toward integrating nuclear and electrolytic technologies for a future low-carbon hydrogen economy.

Table 2—Existing and Planned Nuclear Plants and Infrastructure as per 2025 (Bocca *et al.*, 2023; Olawuyi and Aryanpour, 2024; Shabaneh *et al.*, 2024; Terrapon-Pfaff *et al.*, 2024)

Country	Existing Infrastructure	Major Planned/ Announced Infrastructure	Existing Nuclear Power Plants & Electrolysers	Planned Nuclear Power Plants & Electrolysers	Notes on Relevance to Pink Hydrogen
Qatar	- 800MW Al Kharsaah solar plant. - Ammonia, LNG, advanced steam/ gas systems. - Water desalination plants.	$1bn blue ammonia with CCS. Ammonia/ methanol upgrades. Clean hydrogen strategy.	None (no nuclear or utility-scale electrolysers)	No confirmed nuclear program; electrolyser pilots possible post-2030	Assets can be adapted for pink H_2 if nuclear is introduced.
UAE	- Hydrogen pilot/ demo projects. - Jebel Ali, KEZAD, Khor Fakkan export ports.	National hydrogen strategy; 2,200km H_2 pipeline. Hydrogen "oases" industrial clusters. Export networks.	Barakah Nuclear Plant: Four 1400 MW units (all operational by 2024). PEM/alkaline electrolyser pilots, e.g., TAQA/ADNOC/ENEC.	Future: Additional Barakah-type reactors under consideration. Large-scale electrolyser clusters in planned "oases."	The only active Gulf nuclear power plant (Barakah); grid and steam integration possible for pink H_2.
Oman	- Hydrogen pilots (Hydrom), utilities. - Electrolyser demos. - Sohar/Duqm ports.	"InfraCo" grid: H_2 pipelines, tank storage, LH_2 export. - 1Mt H_2 by 2030, 3.75Mt by 2040 slated. - Water/ renewable expansions.	No nuclear power plants. Pilot electrolysers under Hydrom and partners.	No near-term nuclear program announced. Large-scale electrolyser deployment planned 2025+.	Oman's hydrogen infra is "nuclear-ready" if plants are built; for now, green/blue H_2 dominates.
Egypt	- Major ammonia/ fertilizer plants. - Suez Canal Economic Zone. - Upgraded grid/renewables.	- 32+ MoUs for clean H_2; 36+ mega projects. - Suez ECZ hydrogen/ammonia exports scale-up.	El Dabaa Nuclear Power Plant (under construction): 4x1200 MW, Unit 1 online 2026 (target). Electrolyser pilots in Suez, Ain Sokhna.	Additional Dabaa units possible after 2030. Scaling to multi-100MW electrolysers as MoUs mature.	Egypt will have first operational nuclear grid in Arab Africa (2026+), supporting future pink H_2.
Saudi Arabia	- Aramco/SABIC chemical infrastructure. - Sudair 1.5GW solar. - Early H_2 pilots & blue/green ammonia.	- $30B+ H_2 investment; 4GW NEOM Helios plant. - 11Mt blue ammonia by 2030. - Hydrogen export ports (NEOM, Jazan).	No commercial nuclear plants yet. Small R&D reactors, pilot green/blue ammonia.	2x1400MW nuclear power plants under licensing/FEED (target 2030s). Aramco and partners pursuing gigawatt electrolysers in NEOM/Jazan.	NEOM Helios (4GW) will integrate new large-scale electrolysers. Nuclear will bolster clean H_2 mid/late 2030s.

Safety and Operational Considerations

- Integration of electrolysis systems with nuclear plants requires careful attention to safety protocols and operational procedures (IAEA, 2020). Key considerations include tritium management, chemical compatibility, and emergency response procedures (IAEA, 2020). Impact assessments are required to further investigate other permutations of integration between nuclear and electrolysis technology (Diaz-Pescador *et al.*, 2025).
- Recent research emphasizes the importance of passive safety systems and walk-away safe designs for nuclear-hydrogen integrated facilities (GAIN, 2025). Molten salt reactors particularly excel in this regard due to their inherent safety characteristics and low-pressure operation (GAIN, 2025).

- Load-following capabilities of modern nuclear plants enhance their suitability for hydrogen production, allowing operators to optimize between electricity generation and hydrogen production based on grid demand and market conditions (World Nuclear News, 2024; Innovation News Network, 2025).

Limitations

As mentioned earlier, the Middle East is at the crossroad for developing a comprehensive and divested energy production and transportation program. Regardless of all types of policies and investments being made, it is required to understand the possible cross-cutting challenges in the Middle East that can hinder the Pink Hydrogen Revolution. Cross-cutting challenges are summarized in Table 3 below

Table 3—Cross-Cutting Challenges in MENA

#	Challenge	Middle-East specific manifestation	Key peer-reviewed evidence
1	Desert operating conditions – dust, salts and extreme heat	Wind-blown dust across the Gulf can foul balance-of-plant filters, contaminate feed-water and accelerate electrode ageing, while hot, humid evenings raise stack cooling demand (Becker *et al.*, 2023; Jish Prakash *et al.*, 2015).	Frequent spring dust storms over the Arabian Peninsula are documented by Kalenderski & Stenchikov (2015). Their implications for electrolyser contamination and O&M are analysed in Becker *et al.* (2023).
2	Technology durability & TRL gaps (especially HTSE)	TRL for high-temperature steam electrolysis (HTSE) at commercial scale is still 3–4 in the Gulf; accelerated degradation occurs when locally available brackish or seawater is used without full desalination (Liu *et al.*, 2021).	Liu *et al.* (2021) report rapid chromium volatilisation and seal failures in SOECs tested on Gulf-simulated seawater feeds.
3	High cost of capital (WACC) → elevated LCOH	Gulf sovereign entities can tap low-interest debt, but private developers still face >8 % real WACC because of nascent hydrogen offtake contracts and policy uncertainty, keeping pink-H_2 LCOH above 2.2 US $/kg (Hoogsteyn *et al.*, 2025).	Policy-support sensitivity is quantified in Hoogsteyn *et al.* (2025), showing LCOH falls by 25 – 35 % when contract-for-difference or tax-credit instruments halve the effective WACC.
4	Pipeline & storage readiness	Retrofitting high-pressure gas lines for H_2/N_2 purge and switching to pure H_2 raises embrittlement risk in X60–X70 steels widely used in GCC trunklines. Present TRL is 4–5 (Messaoudani *et al.*, 2016).	The critical review by Messaoudani *et al.* (2016) details fracture-mechanics data showing up to 40 % strength loss at 20 % H_2 by volume, underscoring the retrofit challenge.
5	Social license & regulatory clarity	Recent polling shows strong support for "clean hydrogen", but attitudes towards nuclear-powered production are ambivalent, particularly outside the UAE; misperceptions about radiological risk persist (Alnaqbi and Alami, 2023).	A UAE case-study on public views of low-carbon energy by Alnaqbi & Alami (2023) finds only 54 % of respondents "fully comfortable" with nuclear–hydrogen coupling, confirming the need for targeted engagement campaigns.

Challenges

The pursuit of pink hydrogen, produced via water electrolysis powered by nuclear energy, presents a promising, low-carbon energy vector. However, its widespread adoption, particularly in the Gulf Cooperation Council (GCC) countries, is contingent upon overcoming a series of multifaceted complications. This section delves into the primary challenges, encompassing integration and load management, public perception, regulatory frameworks, and broader technical and economic risks. Endeavor for bringing pink hydrogen into MENA would require four major aspects to be addressed and understood: Economic and Efficiency considerations, Existing Infrastructure, Safety and Operational Consideration, and cross-cutting challenges of region. MENA region controls nearly 150 billion USD of capital investments in Energy market in 2025 alone (IEA, 2025). With developing research into renewables and existing substantial infrastructure of hydrocarbon distribution and developing policies in favor of diversification in the energy grid, Pink Hydrogen development is viable and profitable in the region when all considerations are addressed.

Technical and Infrastructure limitations

Technical Gaps. While electrolysis is an established technology, technical hurdles remain. Advancements are needed to improve the efficiency and reduce the capital cost of electrolysers. For high-temperature electrolysis (HTE), which can offer greater efficiency by using the heat from nuclear reactors, materials science challenges must be overcome as high temperatures can degrade membranes. The development of next-generation reactors, such as Small Modular Reactors (SMRs), is seen as a key enabler for more efficient and flexible pink hydrogen production (Fernández-Arias *et al.*, 2024).

Infrastructure Limitations. A significant bottleneck is the lack of infrastructure for hydrogen transport and storage. Hydrogen's low density makes it costly and challenging to move and store. While blending hydrogen into existing natural gas pipelines is a potential short-term solution, it can cause embrittlement in many current pipelines. A full-scale hydrogen economy will require massive investment in dedicated new pipelines, as well as storage facilities like salt caverns. For exports from the GCC, converting hydrogen to ammonia for shipping is seen as the most viable near-term option.

Integration Strategy and Load Management. A significant hurdle for pink hydrogen production is the effective integration of electrolysis facilities with nuclear power plants. This involves complex strategic decisions and technical coordination.

- **Co-location and Infrastructure.** A key decision is whether to co-locate hydrogen production facilities with nuclear plants or situate them closer to end-users. While transporting electricity is often more economical than transporting hydrogen, on-site production at the nuclear plant guarantees a dedicated, carbon-free electricity source, avoiding the carbon intensity of grid electricity that may include fossil fuels. Co-locating also leverages existing infrastructure and allows for streamlined regulatory oversight in areas already accustomed to stringent safety protocols.
- **Load Management and Hybrid Systems.** Nuclear power plants are designed for continuous, baseload operation, which offers a key advantage over intermittent renewables for hydrogen production. This steady output can drive electrolysis at full capacity around the clock, yielding a stable hydrogen supply (World Nuclear Association, 2024). Furthermore, integrating hydrogen production can provide a flexible load, allowing nuclear plants to maintain constant output while diverting surplus electricity to electrolysis during periods of low grid demand. This enhances the economic case for nuclear power by creating a new revenue stream. However, managing this integration, especially in hybrid systems that might also incorporate variable renewables, is a complex undertaking. Efficient power management strategies are crucial to balance the different energy sources and storage systems.

Public Perception and Social Acceptance

The connection to nuclear energy is a primary barrier to public acceptance of pink hydrogen.

- **Nuclear Stigma.** High-profile nuclear accidents like Chernobyl and Fukushima have created a persistent negative public perception of nuclear technology, raising concerns about safety and radioactive waste management (Fernández-Arias et al. 2024). Overcoming this stigma is crucial for gaining the community support necessary for new pink hydrogen projects. In the Gulf region, this perception has led to nuclear hesitancy; for instance, the Fukushima crisis contributed to the halting of nascent nuclear plans in several countries (The Belfer Center for Science and International Affairs, 2022). Conversely, in the United Arab Emirates (UAE), transparency campaigns and educational outreach have resulted in high public approval for the Barakah nuclear program (Emirates Nuclear Energy Corporation, 2023).

- **Unfamiliarity with Hydrogen.** Hydrogen as a widespread energy carrier is still unfamiliar to the general public. This can make it susceptible to misinformation and negative perceptions shaped by singular events. Educational initiatives are needed to inform the public about the safety and benefits of hydrogen technology.

Economic and Efficiency Considerations

The economic viability of pink hydrogen is a major concern. Key factors influencing its cost-competitiveness include the high initial capital costs of both nuclear plants and electrolysis facilities. Current estimates for pink hydrogen production range from USD 3/kg to USD 5/kg (Association, 2023). By comparison, green hydrogen is in a similar range, while blue hydrogen (from natural gas with carbon capture) can be cheaper depending on gas prices. However, due to the high capacity factors of nuclear power (around 90% compared to 20 to 40% for renewables), pink hydrogen has the potential to be more cost-effective than green hydrogen (IAEA, 2019). Government subsidies and carbon pricing are considered essential to make it competitive with fossil fuels.

- Economic analysis indicates that the cost of electricity, rather than reactor temperature, is the primary driver of nuclear hydrogen production economics (Bass *et al.*, 2020). HTSE requires approximately 40.3 kWhe/kgH_2 for low-temperature reactors versus 38.7 kWhe/kgH_2 for high-temperature reactors - only a 4% reduction (Bass *et al.*, 2020).
- Studies by the National Nuclear Laboratory (NNL) project hydrogen production costs as low as £0.89/kg using thermochemical processes with Advanced Modular Reactors (National Nuclear Laboratory, 2022). The HyTN project demonstrates potential for marked efficiency increases compared to existing hydrogen production technologies (National Nuclear Laboratory, 2022).
- Countries such as India, Russia and China which are part of OECD have nuclear energy costs with 3% discount rate at 27 to 44 $/MWh with the most expensive being Slovakia. Similar rates have been observed globally (Association, 2023).
- IAEA economic evaluation programs indicate that nuclear hydrogen production can be competitive with conventional methods, particularly when considering carbon pricing and grid flexibility services (AGENCY, 2024). The combination of baseload nuclear operation with hydrogen production offers additional revenue streams for nuclear plant operators (Hjelmeland *et al.*, 2025).

Regulations in the GCC Countries

The GCC region, with significant investments in both nuclear energy and hydrogen, faces the challenge of developing a comprehensive regulatory landscape. Globally, a lack of clear and consistent policies can create uncertainty for investors. Establishing international standards for what qualifies as "clean" hydrogen is crucial. Furthermore, the geopolitical landscape of the Middle East presents unique risks, as nuclear facilities can be potential military targets (The Belfer Center for Science and International Affairs, 2022)

- **Nascent Regulatory Frameworks.** While countries like the UAE, Saudi Arabia, and Oman have announced national hydrogen strategies, the specific legal and policy frameworks to govern the entire hydrogen value chain are largely in early development. There is currently no unified hydrogen code in the MENA region, and regulations for production, transport, and trade remain to be fully formulated (Olawuyi and Aryanpour, 2024). Comprehensive safety codes and certification systems are also lacking. The UAE has been proactive, with Abu Dhabi's Department of Energy developing a hydrogen policy to accelerate the national strategy (Norton Rose Fulbright, 2024).
- **Need for Harmonization and Incentives.** To attract foreign investment and technology, GCC nations need to establish clear, harmonised regulations and standards. This includes developing

robust licensing processes and health and safety standards. Financial incentives are also seen as crucial to stimulate both production and consumption, bridging the economic gap until the market matures. At a global level, agreements on the mutual recognition of "low-carbon hydrogen," such as the one made at COP28, are vital for enabling trade with regions like Europe (COP28, 2023).

It is important as well to note that the region hosts some of the world's most ambitious nuclear programs:

- **UAE**: Four APR-1400 units at **Barakah** (5.6 GW_e) supply ≈ 25 % of national power and displace 21 Mt CO_2-eq yr^{-1} (ENEC 2024).
- **Saudi Arabia**: A 2.8 GW two-unit pressurized-water-reactor (PWR) tender is at preferred-bidder stage; SMR options for remote mining sites are under review (CSIS 2023).
- **Egypt**: Construction of four 1,200 MW VVER-1200 units at **El-Dabaa** began in 2022; grid connection is expected 2029–2032 (Rosatom 2025).
- **Qatar** and **Oman**: Neither operate reactors, but both have SMR feasibility studies—Qatar for desalination-coupled plants at Ras Laffan (Earthna 2025) and Oman for the Duqm free-zone (OHC 2024).

The technology maturity to implement such type of integrated system is quite high and promising due to which above ambitious projects are currently underway (Beck and Pincock, 2011; BioEnergyTimes, 2025; Brauns and Turek, 2020; Du *et al.*, 2025; Nasser and Hassan, 2024; Razi and Dincer, 2022). Integration strategies are also to be considered which can also delve into integrating different hydrogen hubs together to increase the production capacity (Kim *et al.*, 2024; Yoo *et al.*, 2025).

Case Study: Al Barakah Nuclear Power Plant – UAE

Scope and objective

Al Barakah Nuclear power plant is the largest source and the first nuclear power plant in the Arab world. It's a significant milestone for both the UAE and South Korea, which supplied the reactor technology. The plant features four APR-1400 reactors with a combined capacity of 5,600 MW, designed to provide up to 28% of the UAE's electricity needs. Construction on the plant began in 2009, with the first reactor going online in 2020. This section provides a step-by-step, research-style extraction of the technical and economic feasibility assessment for system-level Levelised Cost of Hydrogen (LCOH) in the United Arab Emirates. Electricity from the Barakah nuclear fleet is allocated to large-scale electrolysis, and four configuration options are screened. This section utilizes the current established nuclear power plant and combines with a suitable Electrolyser technology to determine LCOH in order to define which integrated hydrogen production system is viable. The analysis explicitly models the complete electrical chain from the nuclear plant switchyard to the electrolyser direct-current bus, together with balance-of-plant utilities (cooling, de-ionised water) and periodic stack refurbishment. The decision indicator is the system-level LCOH (USD/kg).

Configurations assessed:

1. Configuration 1: Co-located, process-integrated Solid Oxide Electrolyser Cell (SOEC) with steam and cooling integration at the nuclear site.
2. Configuration 2: Co-located, power-only alkaline electrolysis with standalone balance-of-plant at the nuclear site.
3. Configuration 3: Remote electrolysis plant located 53 kilometers from the nuclear plant, supplied by High Voltage Alternating Current (HVAC) transmission.
4. Configuration 4: Remote electrolysis plant located 53 kilometers from the nuclear plant, supplied by High Voltage Direct Current (HVDC) transmission.

Basis of assumptions

Following assumptions in Table 4 are utilized with governing equations which include base costs and energy requirements for a typical integrated nuclear powered hydrogen production system(Alnaqbi and Alami, 2023; Arcos and Santos, 2023; IEA, 2025).

Table 4—Costs and Assumptions (Alnaqbi and Alami, 2023; Arcos and Santos, 2023; IEA, 2025).

Parameter	Cost
Power Allocation	90% Utilization – 6,400 hours per annum Gross Energy at Nuclear Busbar E_{gross} = 10,752,000 MWh/yr
Electrolyser specific energy consumption (SEC)	SOEC = 40 kWh/ kg of Hydrogen (with steam/heat integration) Alkaline Electrolysis = 55 kWh/ kg of Hydrogen (electricity only)
Electrolyser and balance-of-plant capital expenditure	SOEC = 1,200 USD per kW (USD 2.016 Billion) Alkaline Electrolysis = 800 USD per kW (USD 1.344 Billion)
Electrical/transmission capital expenditure	Co-located electricals = USD 150 million Remote HVAC (53 km) = USD 520 million Remote HVDC (53 km) = USD 1.06 billion.
Process Integration Allowance (Configuration 1 only)	USD 100 million for steam extraction/condensate return, heat exchangers, process piping, control integration, and cooling tie-ins.
Fixed operation and maintenance	Electrolyser 2% of electrolyser capital expenditure per annum Transmission/electrical 1.5% of the corresponding capital expenditure per annum Process integration 1.5% per annum.
Refurbishment allowance (annualised)	SOEC 12% of electrolyser capital expenditure per annum Alkaline 3% per annum
Economic life and financing:	25 years *Real Weighted Average Cost of Capital = 8%; Capital Recovery Factor = r(1+r)^n / [(1+r)^n − 1] = 0.09368.*
Electricity price	Levelised Cost of Electricity at the nuclear switchyard (charged on gross withdrawal) = USD 80 per megawatt-hour.
Losses and parasitic consumption	HV transformer 0.5%; HVAC line 1.8% (53 km); HVDC line 0.16% + converters 2.0%; MV transformer 0.7%; rectifier efficiency 98%; auxiliaries 1.5% (co-located) or 2.0% (remote).
Water and utilities:	de-ionised water 9 litres per kilogram; utilities and consumables adder USD 0.05 per kilogram of hydrogen.

Variable definitions and governing equations

Symbols [units]:

- E_{gross} [MWh/a]: annual gross electrical energy at the nuclear busbar
- η_{chain} [–]: end-to-end efficiency from busbar to rectifier direct-current output (excl. auxiliaries).
- E_{aux} [MWh/a]: auxiliary consumption (fraction of E_{gross}).
- $E_{to\text{-}stacks}$ [MWh/a]: net direct-current energy delivered to electrolyser stacks.
- SEC [MWh/kg]: specific electrical energy per kilogram of hydrogen.
- H_{annual} [kg/a]: annual hydrogen production.
- $CAPEX_{elec}$ [USD]: electrolyser and balance-of-plant capital expenditure.
- $CAPEX_{T\&D}$ [USD]: transmission and electrical balance capital expenditure.
- $CAPEX_{PIA}$ [USD]: process-integration allowance (Configuration 1).
- CRF [–]: capital recovery factor derived from Weighted Average Cost of Capital and life.

- C_{total} [USD/a]: total annual cost; LCOH_system [USD/kg]: C_{total} divided by H_{annual}.

Equations:

$$\eta_{chain}(\text{colocated}) = (1-\ell_{HV}) \times (1-\ell_{MV}) \times \eta_{rectifier} \quad \text{Eq (1)}$$

$$\eta_{chain}(\text{remote-HVAC}) = (1-\ell_{HV}) \times (1-\ell_{HVAC,line}) \times (1-\ell_{MV}) \times \eta_{rectifier} \quad \text{Eq (2)}$$

$$\eta_{chain}(\text{remote-HVDC}) = (1-\ell_{HV}) \times (1-\ell_{conv}) \times (1-\ell_{HVDC,line}) \times (1-\ell_{MV}) \times \eta_{rectifier} \quad \text{Eq (3)}$$

$$E_{to\text{-}stacks} = E_{gross} \times \eta_{chain} - E_{aux} \quad \text{Eq (4)}$$

$$H_{annual} = E_{to\text{-}stacks} / SEC \quad \text{Eq (5)}$$

$$C_{electricity} = E_{gross} \times LCOE \quad \text{Eq (6)}$$

$$C_{cap,elec} = CRF \times CAPEX_{elec}; C_{O\&M,elec} = 0.02 \times CAPEX_{elec}; C_{refurb} = f_{refurb} \times CAPEX_{elec} \quad \text{Eq (7)}$$

$$C_{cap,T\&D} = CRF \times CAPEX_{T\&D}; C_{O\&M,T\&D} = 0.015 \times CAPEX_{T\&D} \quad \text{Eq (8)}$$

$$C_{cap,PIA} = CRF \times CAPEX_{PIA}; C_{O\&M,PIA} = 0.015 \times CAPEX_{PIA} (\text{Config. 1 only}) \quad \text{Eq (9)}$$

$$C_{utils} = 0.05 \times H_{annual} (USD/kg \times kg/a) \quad \text{Eq (10)}$$

$$C_{total} = \text{sum of all components above;} \quad \text{Eq (11)}$$

$$LCOH_{system} = C_{total} / H_{annual} \quad \text{Eq (12)}$$

Results

- Common values: E_{gross} = 10,752,000 megawatt-hours per annum.
- $C_{electricity} = E_{gross} \times LCOE$ = 10,752,000 × 80 = 860.16 ×10^6 United States dollars per annum.

Configuration 1 — Co-located, process-integrated SOEC

Loss chain and auxiliaries: ℓ_{HV} = 0.5%, ℓ_{MV} = 0.7%, $\eta_{rectifier}$ = 98%; auxiliaries = 1.5% of E_{gross}.
η_{chain} = (1−0.005) × (1−0.007) × 0.98 = 0.9683 (≈ 96.83%).
$E_{to\text{-}stacks}$ = 10.752 TWh × 0.9683 − 0.015 × 10.752 TWh = 10.2496 TWh/y.
H_{annual} = 10.2496 TWh ÷ 0.040 MWh/kg = 256.240 ×10^6 kg/a (256,240 t/a).
Annualised cost components (million USD per annum):

Table 5—Configuration 1 Costing Summary

Parameter	Cost	Unit
$C_{electricity}$	860.16	MUSD/y
$C_{cap,elec}$	188.86	MUSD/y
$C_{O\&M,elec}$	40.32	MUSD/y
C_{refurb}	241.92	MUSD/y
$C_{cap,T\&D}$	14.05	MUSD/y
$C_{O\&M,T\&D}$	2.25	MUSD/y
$C_{cap,PIA}$	9.37	MUSD/y
$C_{O\&M,PIA}$	1.5	MUSD/y
C_{utils}	12.81	MUSD/y

Parameter	Cost	Unit
C_{total}	1.37 Billion USD	
LCOH	5.35 USD/kg	

Configuration 2 — Co-located, power-only alkaline electrolysis

Loss chain and auxiliaries: ℓ_{HV} = 0.5%, ℓ_{MV} = 0.7%, $\eta_{rectifier}$ = 98%; auxiliaries = 1.5% of E_{gross}.
$\eta_{chain} = (1-0.005) \times (1-0.007) \times 0.98 = 0.9683$ (≈ 96.83%).
$E_{to\text{-}stacks} = 10.752$ TWh × 0.9683 − 0.015 × 10.752 TWh = 10.2496 TWh/y.
$H_{annual} = 10.2496$ TWh ÷ 0.055 MWh/kg = 186.356×10^6 kg/a (186,356 t/a).
Annualised cost components (million USD per annum):

Table 6—Configuration 2 Costing Summary

Parameter	Cost	Unit
$C_{electricity}$	860.16	MUSD/y
$C_{cap,elec}$	125.91	MUSD/y
$C_{O\&M,elec}$	26.88	MUSD/y
C_{refurb}	40.32	MUSD/y
$C_{cap,T\&D}$	14.05	MUSD/y
$C_{O\&M,T\&D}$	2.25	MUSD/y
$C_{cap,PIA}$	0.0	MUSD/y
$C_{O\&M,PIA}$	0.0	MUSD/y
C_{utils}	9.32	MUSD/y
C_{total}	1.079 Billion USD	
LCOH	5.79 USD/kg	

Configuration 3 — Remote 53 km, HVAC transmission, alkaline electrolysis

Loss chain and auxiliaries: ℓ_{HV} = 0.5%, ℓ_{MV} = 0.7%, $\eta_{rectifier}$ = 98%; auxiliaries = 2.0% of E_{gross}.
$\eta_{chain} = (1-0.005) \times (1-0.007) \times (1-0.018) \times 0.98 = 0.9508$ (≈ 95.08%).
$E_{to\text{-}stacks} = 10.752$ TWh × 0.9508 − 0.020 × 10.752 TWh = 10.0084 TWh/y.
$H_{annual} = 10.0084$ TWh ÷ 0.055 MWh/kg = 181.972×10^6 kg/a (181,972 t/a).
Annualised cost components (million USD per annum):

Table 7—Configuration 3 Costing Summary

Parameter	Cost	Unit
$C_{electricity}$	860.16	MUSD/y
$C_{cap,elec}$	125.91	MUSD/y
$C_{O\&M,elec}$	26.88	MUSD/y
C_{refurb}	40.32	MUSD/y
$C_{cap,T\&D}$	48.71	MUSD/y
$C_{O\&M,T\&D}$	7.80	MUSD/y
$C_{cap,PIA}$	0.0	MUSD/y

Parameter	Cost	Unit
$C_{O\&M,PIA}$	0.0	MUSD/y
C_{utils}	9.10	MUSD/y
C_{total}	1.119 Billion USD	
LCOH	6.16 USD/kg	

Configuration 4 — Remote 53 km, HVDC transmission, alkaline electrolysis

Loss chain and auxiliaries: ℓ_{HV} = 0.5%, converters (pair) = 2.0%, $\ell_{HVDC,line}$ = 0.16%, ℓ_{MV} = 0.7%, $\eta_{rectifier}$ = 98%; auxiliaries = 2.0% of E_{gross}.
$\eta_{chain} = (1-0.005) \times (1-0.020) \times (1-0.0016) \times 0.98 = 0.9474$ (≈ 94.74%).
$E_{to\text{-}stacks} = 10.752$ TWh × 0.9474 − 0.020 × 10.752 TWh = 9.9713 TWh/y.
$H_{annual} = 9.9713$ TWh ÷ 0.055 MWh/kg = 181.296×10^6 kg/a (181,296 t/a).
Annualised cost components (million USD per annum):

Table 8—Configuration 4 Costing Summary

Parameter	Cost	Unit
$C_{electricity}$	860.16	MUSD/y
$C_{cap,elec}$	125.91	MUSD/y
$C_{O\&M,elec}$	26.88	MUSD/y
C_{refurb}	40.32	MUSD/y
$C_{cap,T\&D}$	99.30	MUSD/y
$C_{O\&M,T\&D}$	15.90	MUSD/y
$C_{cap,PIA}$	0.0	MUSD/y
$C_{O\&M,PIA}$	0.0	MUSD/y
C_{utils}	9.06	MUSD/y
C_{total}	1.178 Billion USD	
LCOH	6.50 USD/kg	

Summary

Based on comparative analysis (Table 9) **Co-located, process-integrated SOEC** (Configuration 1) remains the most suitable configuration under the stated assumptions. It achieves the lowest LCOH owing to (i) reduced specific electrical energy via steam integration, (ii) shared cooling and water systems, and (iii) avoidance of transmission capital expenditure and losses. At a separation of 53 kilometers, High Voltage Alternating Current (Configuration 3) is preferred to High Voltage Direct Current (Configuration 4) because the additional converter station capital expenditure and conversion losses of HVDC dominate short distances.

Table 9—Comparative Analysis

Configuration	Transmission	Technology	H_2 Output (t/a)	Total Annual Cost (Billion USD/a)	LCOH (USD/kg)
C1	Co-located	SOEC	256,240	1.371	5.35
C2	Co-located	Alkaline	186,356	1.079	5.79

Configuration	Transmission	Technology	H_2 Output (t/a)	Total Annual Cost (Billion USD/a)	LCOH (USD/kg)
C3	HVAC (53 km)	Alkaline	181,972	1.119	6.15
C4	HVDC (53 km)	Alkaline	181,296	1.178	6.50

Benchmarking against blue hydrogen

For context, the best pink hydrogen configuration (Configuration 1) at 5.35 USD/kg is benchmarked (Table 10) against indicative blue hydrogen delivered-cost baselines (USD 2.5–3.5/kg; and a high case of USD 5.0/kg reflecting elevated gas and full chain carbon transport and storage tariffs).

Table 10—Benchmarking analysis against blue hydrogen LCOH

Blue H_2 Baseline (USD/kg)	Pink H_2 (C1) LCOH (USD/kg)	Δ (USD/kg)	Δ (%)
2.50- 5	5.35	+2.85-	+114.1%-7%
3.00	5.35	+2.35	+78.4%
3.50	5.35	+1.85	+52.9%
5.00	5.35	+0.35	+7.0%

Relative to USD 2.5–3.5/kg blue hydrogen, pink gold hydrogen at the current assumptions is higher by approximately 53–114%; at USD 5.0/kg blue hydrogen it is approximately 7% higher. The principal levers to narrow the gap are electricity tariff, cost of capital, and stack refurbishment intensity.

Sensitivity and uncertainty

Utilizing the data shown in tables above, sensitivity check was considered in order to determine the weightage of each variable utilized in assessment of LCOH. Following table 11 includes the analysis data which was assessed during sensitivity analysis and its complete impact on LCOH.

Table 11—Data for sensitivity analysis

Parameter	Low Value	Base Value	High Value	Low Impact (USD/kg)	High Impact (USD/kg)	Total Impact
Electricity Price (USD/MWh)	70	80	90	-2.85	2.85	5.7
Refurb Rate (%)	8	12	16	1.4	-1.4	2.8
Capacity Factor (%)	95	90	85	-0.2	0.2	0.4
Eletrolyser CAPEX Reduction (%)	30	0	-10	-0.24	0.08	0.32
WACC (%)	4	8	12	0.03	-0.03	0.06

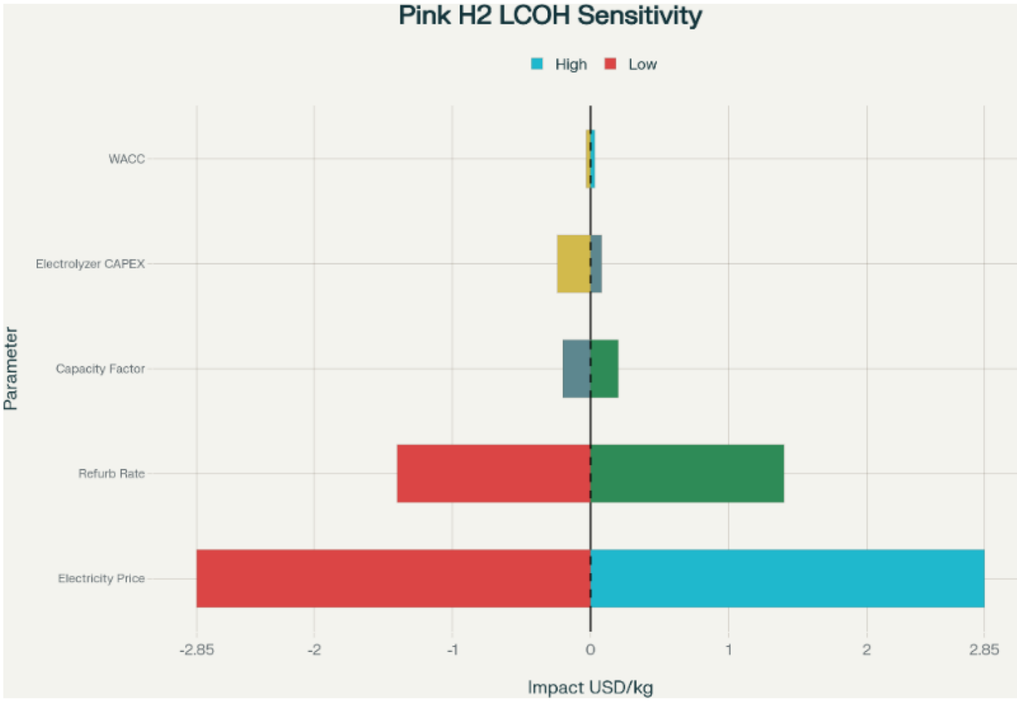

Figure 9—Tornado Plot – Pink H_2 LCOH

Based on the above data, Electricity prices has the most impact on LCOH.

- Electricity tariff sensitivity: $LCOH_{system}$ varies approximately linearly with LCOE. At this scale, a ±10 USD/MWh change in LCOE shifts LCOH by ≈ ±0.27–0.30 USD/kg. Financing sensitivity: reducing the real Weighted Average Cost of Capital from 8% to 4% (Capital Recovery Factor ≈ 0.064) lowers LCOH by ~12–15%, with the largest absolute impact in capital-intensive configurations. Refurbishment sensitivity: reducing the SOEC refurbishment allowance from 12% to 8% of electrolyser capital expenditure per annum improves LCOH by ~0.30–0.40 USD/kg.
  - South Korea Experience: The Korean nuclear program achieved ~40% cost reduction through continuous construction and standardization, with domestic costs at \$3,571/kW compared to Barakah export costs of \$5,947/kW.
  - SMR Learning Curves: Studies show potential for 35-46% cost reductions from First-of-a-Kind (FOAK) to Nth-of-a-Kind (NOAK) deployment, with factory production enabling 30-50% reductions.
  - Nuclear Learning Rates: Academic reviews show highly variable learning rates (-38% to +15%), but successful programs with continuous construction demonstrate positive learning effects (Rubin *et al.*, 2015).
- Uncertainty classification: The estimate is prepared at order-of-magnitude level using capacity-factored and parametric methods. In line with AACE International Class 5 (Class V) practice, accuracy is typically −50% to +100%, subject to scope definition and data quality. Subsequent phases shall refine quantities and vendor pricing and include indirect costs and contingencies as appropriate.

Roadmap and Recommendations

The Gulf Cooperation Council (GCC) states, together with Egypt, are positioned to become global leaders in pink hydrogen production—hydrogen generated through nuclear-powered electrolysis. This comprehensive policy roadmap outlines a strategic framework across five critical pillars to facilitate the development of a thriving pink hydrogen economy by 2031, requiring a total investment of $66.4 billion.

Table 12—Five-Point Action Plan – Pink Hydrogen Revolution

Pillar	Action	Lead agencies	Timeline
Finance	Sovereign-fund-backed *Hydrogen Innovation Facility* offering 2 % concessional loans for first-of-a-kind projects	ADQ, PIF, QIA, EFIA, OFID, QNB	2026-
Regulation	GCC+Egypt **Pink Hydrogen Charter** adopting uniform GO, safety and water-use standards	GCC-SG, LEC Egypt, NRA Oman	draft 2027, adopt 2028
Manufacturing	Free-zone incentives for electrolyser balance-of-plant (BoP) factories in Abu Dhabi and Dammam	MoIAT UAE, MIM Saudi	2026–30
Infrastructure	Phase-I retrofit of existing gas pipeline Ruwais → Jubail (hydrogen blend 20 %)	ADNOC, Aramco	FID 2027, online 2031
Engagement	Regional public-outreach programme showcasing Barakah, El-Dabaa safety data	IAEA, national regulators	2025-

The following figure encapsulates the major aspects of the five pillars for MENA Pink H_2 Charter which can be responsible in establishing a viable market for Pink H_2. The pillars include a financial roadmap to establish engagement from society in terms of certain cross-cutting challenges.

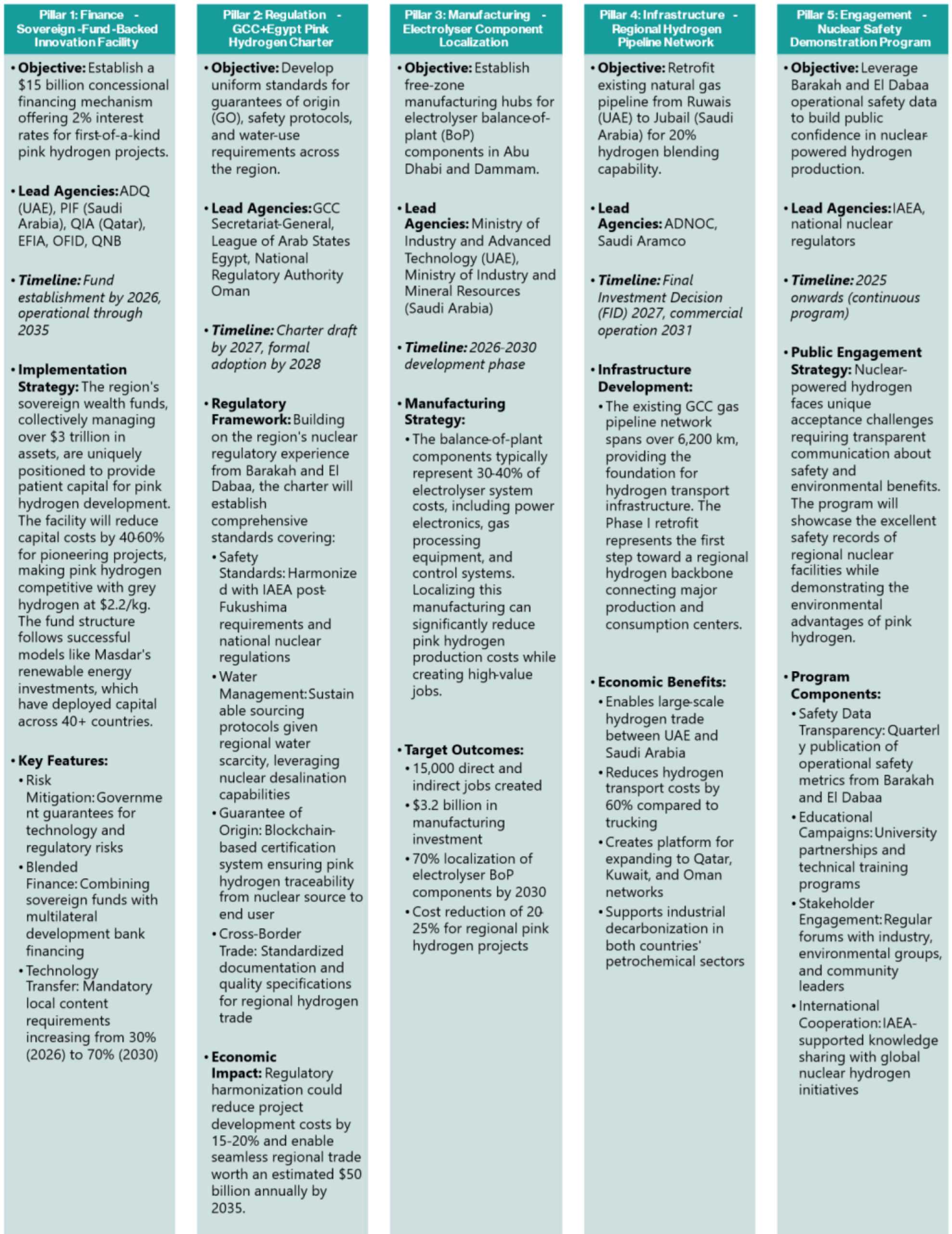

Figure 10—Policy Roadmap – MENA Pillars

Implementation Timeline and Milestones

Figure shows a developed timeline for the proposed Pink Hydrogen Charter based upon the five pillars – Finance, Regulation, Manufacturing, Infrastructure and Deployment. Each pillar represents the action required for reaching a fully developed Pink H_2 market in the MENA region.

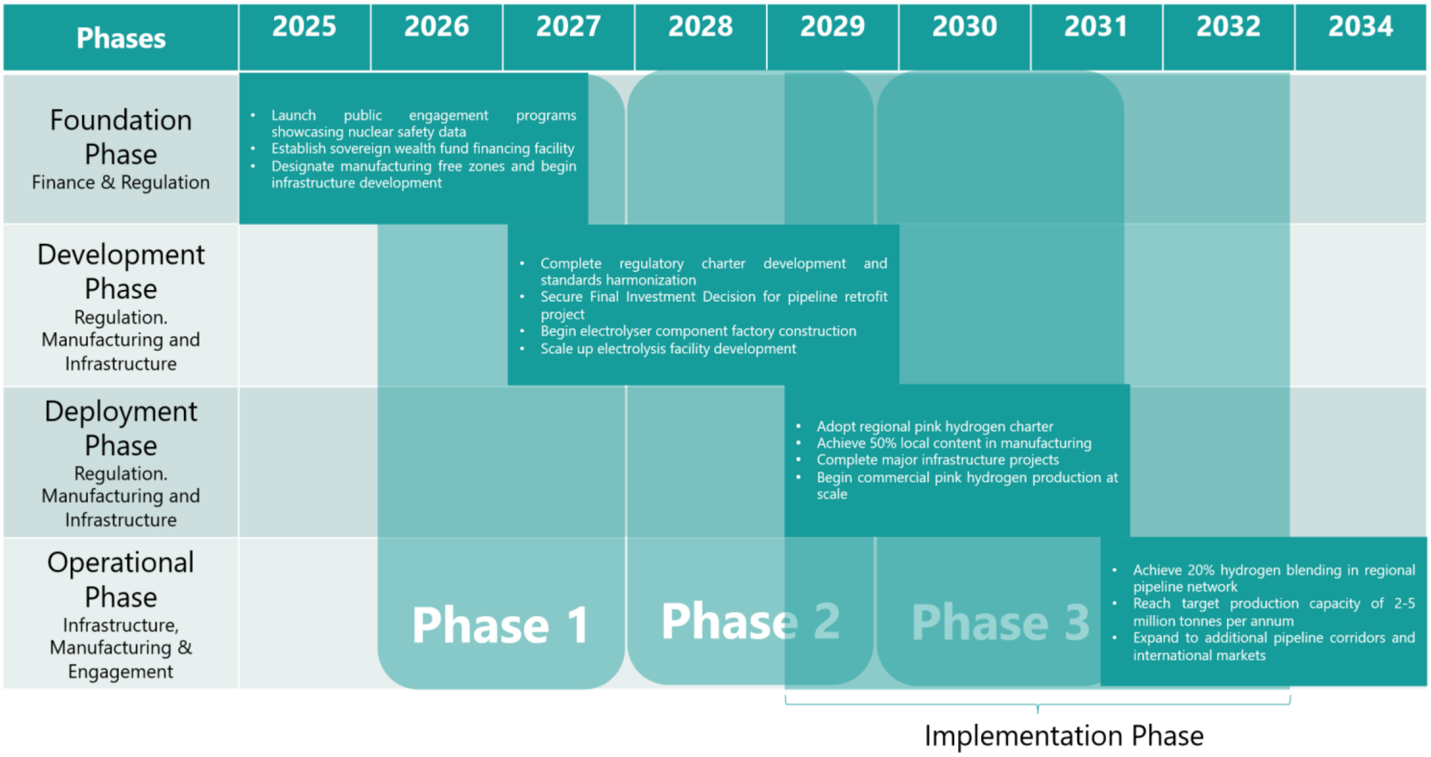

Figure 11—Policy Roadmap – MENA Roadmap

Conclusion and Recommendations

The Middle East and North Africa stand poised at a strategic inflection point as the global energy transition accelerates. This paper demonstrates that nuclear-driven hydrogen. The so-called "pink hydrogen"—offers a technically robust, scalable, and increasingly competitive pathway for producing clean hydrogen at industrial scale. By leveraging the region's nascent and planned nuclear infrastructure and integrating advanced high-temperature steam electrolysis (SOEC) technologies, significant gains in energy efficiency and lifecycle emissions reduction are achievable. Key studies and pilot projects worldwide validate technical feasibility and highlight the unique process heat and baseload power advantages of nuclear hydrogen coupling.

However, deployment across MENA requires careful navigation of regional challenges: harsh desert environments, technology readiness, capital costs, regulatory complexity, and public acceptance. Our detailed techno-economic assessment establishes that process-integrated, co-located nuclear-electrolyser sites (particularly using SOEC) yield the most favorable costs, with the levelized cost of hydrogen reaching 5.35 USD/kg under current scenarios. While this remains higher than blue hydrogen benchmarks at today's gas prices, targeted interventions—particularly lowering electricity tariffs, reducing financing costs, and supporting local electrolyser supply chains—can close the gap.

Looking ahead, strong regional coordination on financing, standards, manufacturing, and pipeline readiness will be essential. The creation of regional hydrogen facilities harmonized regulatory frameworks, and ambitious scale-up plans—while proactively engaging the public—can establish MENA as a global leader in pink hydrogen. Early-mover states such as the UAE and Egypt will set the example, but the entire region has an opportunity to redefine its role in the future of energy through the development of safe, efficient, and economically viable nuclear-powered hydrogen value chains.

In summary, the "pink gold" revolution is within reach, provided that MENA countries seize the opportunity to synergize nuclear and hydrogen strategies, invest in innovation, and build the enabling environment needed for global market leadership in the decarbonized energy era.

Nomenclature

ADIPEC – Abu Dhabi International Petroleum Exhibition & Conference
ADQ – Abu Dhabi Developmental Holding Company
AMR – Advanced Modular Reactor (UK terminology, a class of next-gen reactors)
APR-1400 – Advanced Power Reactor 1400 (MW class, Korean Gen-III nuclear reactor)
AWE – Alkaline Water Electrolysis
CAPEX – Capital Expenditure
COP – Conference of the Parties (UN Climate Change Conference)
COP26 / COP28 / COP29 – Specific UN climate summits held in 2021, 2023, 2024/2025 respectively
CRF – Capital Recovery Factor
CSIS – Center for Strategic & International Studies
DCS – Distributed Control System
DOE – U.S. Department of Energy
EFIA – Egypt Fund for Investment and Advisory (context: finance pillar)
ENEC – Emirates Nuclear Energy Corporation
GAIN – Gateway for Accelerated Innovation in Nuclear (DOE initiative)
GO – Guarantee of Origin (green/pink hydrogen certification context)
H_2 – Hydrogen
HAZID – Hazard Identification (safety methodology)
HTE – High Temperature Electrolysis
HTGR – High Temperature Gas-cooled Reactor
HTSE – High Temperature Steam Electrolysis
HVAC – High Voltage Alternating Current (also common as Heating, Ventilation, Air Conditioning, but here it's grid transmission)
HVDC – High Voltage Direct Current
IAEA – International Atomic Energy Agency
IEA – International Energy Agency
ILS – Integrated Laboratory-Scale (experiment at INL for SOEC testing)
IRENA – International Renewable Energy Agency
kWh/kg – Kilowatt-hour per kilogram
LCOH – Levelized Cost of Hydrogen
LEC Egypt – Likely Local Energy Commission (Egypt, context: regulatory body in hydrogen charter)
MIM Saudi – Ministry of Industry and Mineral Resources, Saudi Arabia
MJ – Megajoule
MoIAT UAE – Ministry of Industry and Advanced Technology, UAE
MV – Medium Voltage
MWh – Megawatt-hour
NNL / UKNNL – (UK) National Nuclear Laboratory
NRA Oman – Nuclear Regulatory Authority (Oman, in charter context)
O&M – Operation and Maintenance
OECD – Organisation for Economic Co-operation and Development
OFID – OPEC Fund for International Development
PCHE – Printed Circuit Heat Exchanger
PEM – Polymer Electrolyte Membrane (or Proton Exchange Membrane, both are used in electrolysis)

PIF – Public Investment Fund (Saudi Arabia)
PWR – Pressurized Water Reactor
QIA – Qatar Investment Authority
QNB – Qatar National Bank
SCFD – Standard Cubic Feet per Day
SEC – Specific Energy Consumption
SIS – Safety Instrumented System
SMR – Small Modular Reactor
SOEC – Solid Oxide Electrolysis Cell
TRL – Technology Readiness Level
VVER-1200 – Water-Water Energetic Reactor (Russian 1200 MW PWR design)
WACC – Weighted Average Cost of Capital

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